

Lecture 3&4

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Fundamental Equation of Thermodynamics

An infinitesimal change in internal energy of a reversible process (from initial state I to final state F) is given by

$$\begin{aligned}dU &= dq - p dV \\dU &= T dS - p dV\end{aligned}\tag{i}$$

Although calculated for a reversible process from state I to F, the above equation is also equally valid for an irreversible process involving the same initial (I) and final states (F), since U is a state function.

Infinitesimal change in enthalpy can be expressed as

$$\begin{aligned}dH &= d(U + PV) \\dH &= dU + p dV + V dp \\dH &= T dS - p dV + p dV + V dp \\dH &= T dS + V dp\end{aligned}\tag{ii}$$

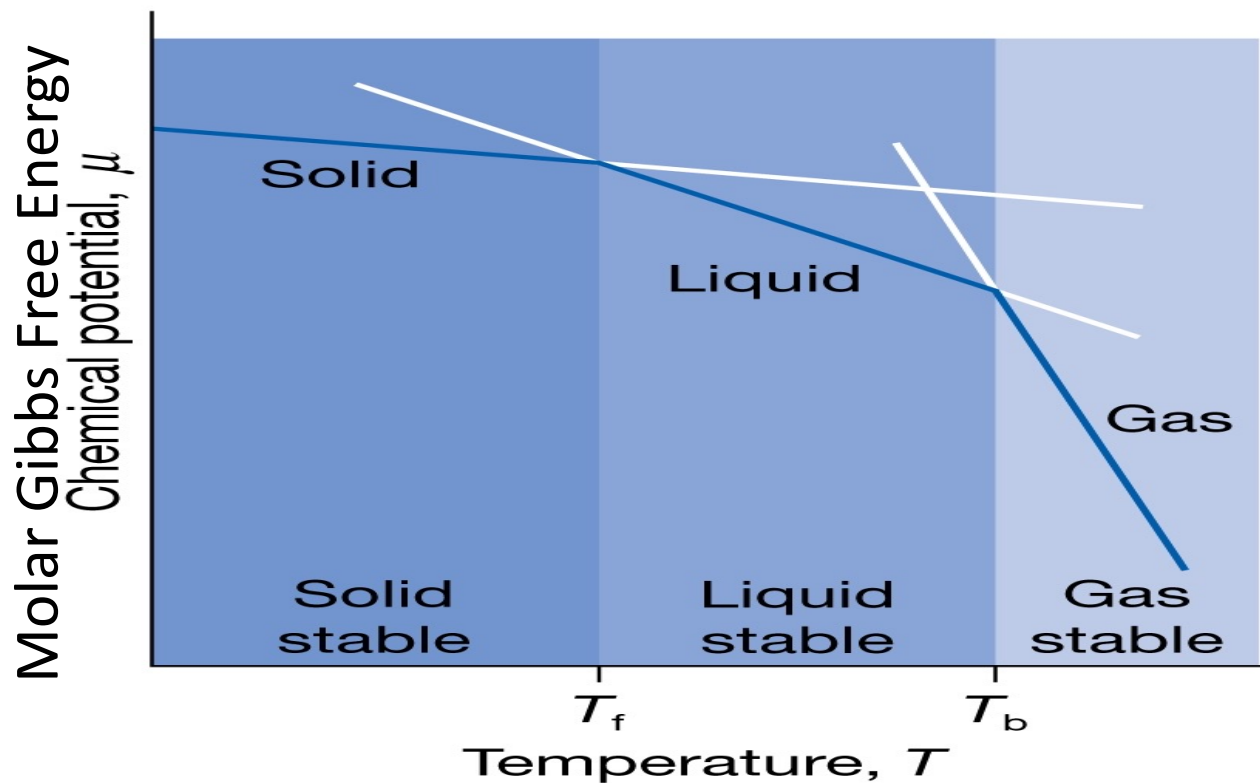
Infinitesimal change in Helmholtz energy can be expressed as

$$\begin{aligned}dA &= d(U - TS) \\dA &= dU - T dS - S dT \\dA &= T dS - p dV - T dS - S dT \\dA &= -S dT - p dV\end{aligned}\tag{iii}$$

Infinitesimal change in Gibbs free energy can be expressed as

$$\begin{aligned}dG &= dH - T dS - S dT \\dG &= T dS + V dp - T dS - S dT \\dG &= V dp - S dT\end{aligned}\tag{iv}$$

These four equations (i-iv) are fundamental equations of thermodynamics.



$$dG = -SdT + Vdp$$

$$\left(\frac{\partial G}{\partial T}\right)_p = -S$$

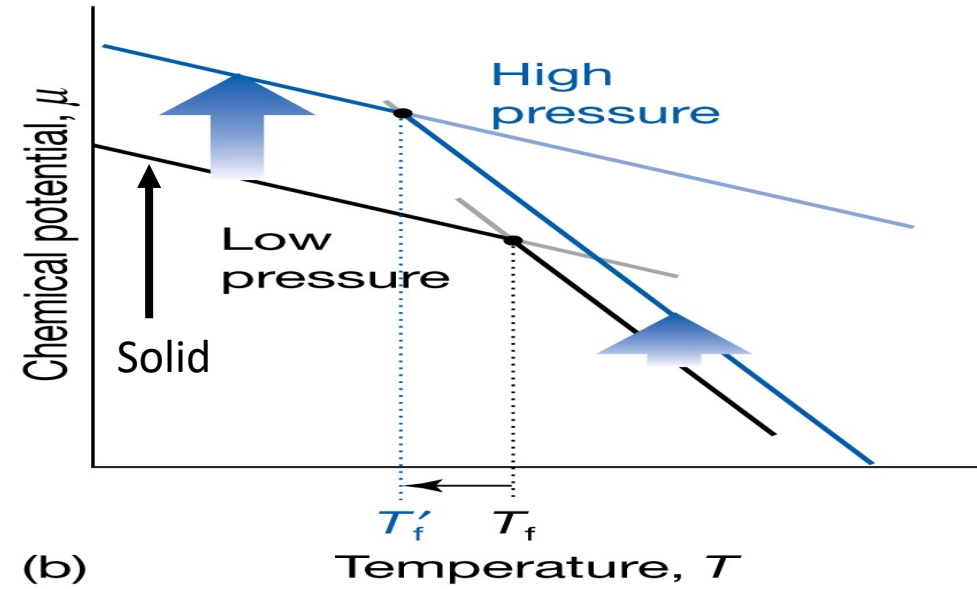
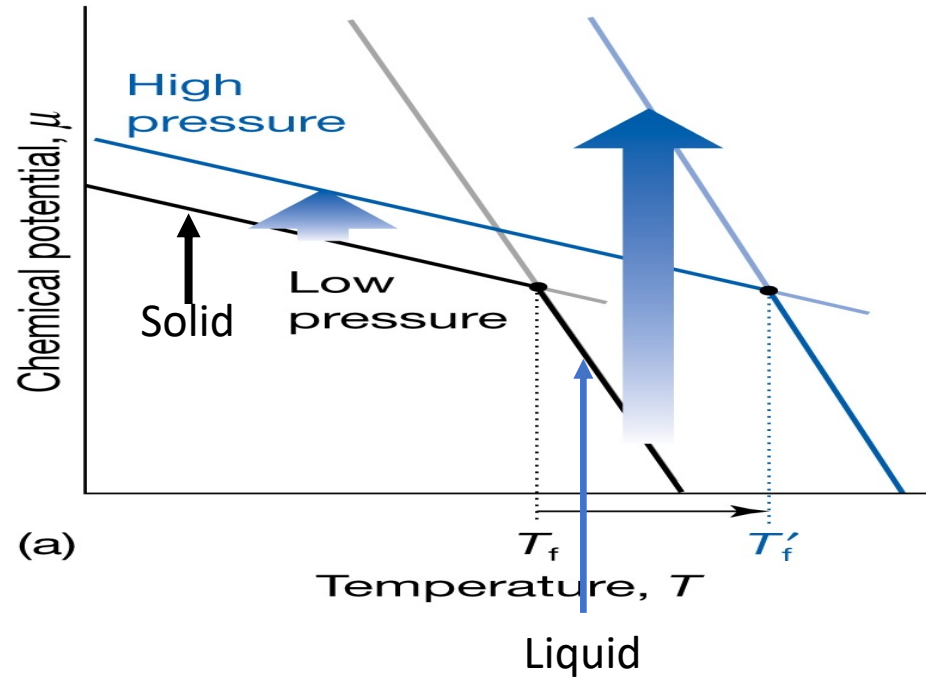
$$\left(\frac{\partial G_m}{\partial T}\right)_p = -S_m$$

$$\left(\frac{\partial \mu}{\partial T}\right)_p = -S_m \quad (\text{negative quantity})$$

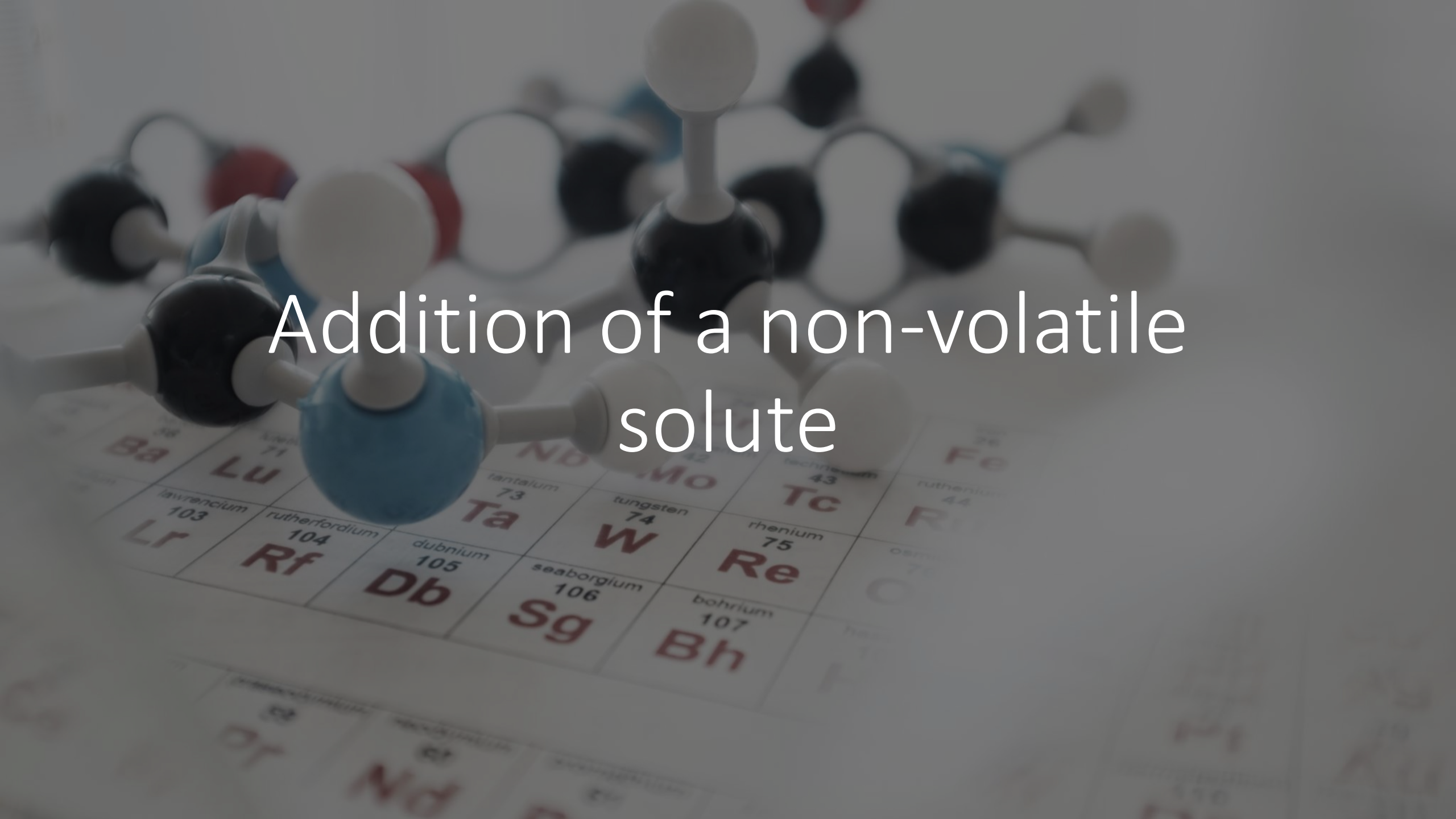
$$S_m(\text{g}) > S_m(\text{l}) > S_m(\text{s})$$

Response of melting temperature to the applied pressure

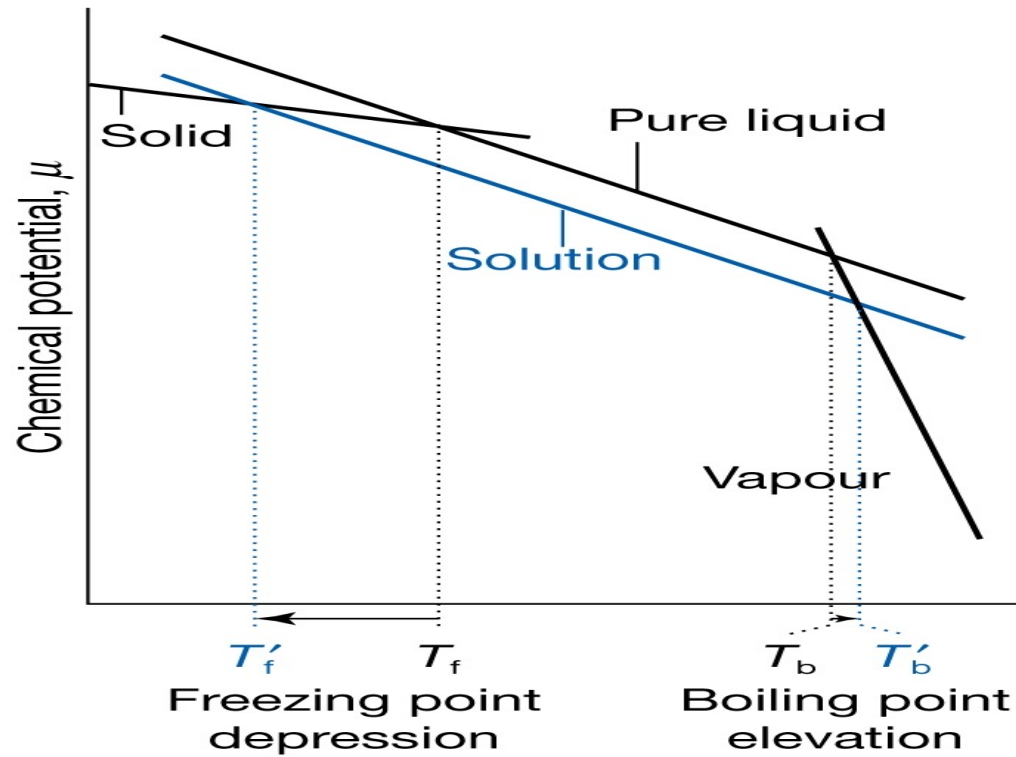
Water



$$\left(\frac{\partial G}{\partial p}\right)_T = V$$



Addition of a non-volatile
solute



There is elevation of boiling point upon addition of a solute.

There is depression of freezing point upon addition of a solute.

A close-up, high-contrast photograph of a mechanical coupling. The image shows a polished metal shaft with a threaded section, likely a coupling or a part of a machine. The lighting is dramatic, with strong highlights and deep shadows, emphasizing the metallic texture and the precision of the manufacturing. The background is blurred, focusing attention on the central mechanical component.

Coupling with Extra work

$$dU = TdS - pdV$$

$$dH = TdS + Vdp$$

$$dA = -SdT - pdV$$

$$dG = Vdp - SdT$$

Are these equations valid for irreversible reaction ?

At constant temperature and pressure, $dG=0$, then how free energy will decrease for a spontaneous process at constant p and T ?

Important Thermodynamic equations

$$dU = TdS - pdV + dw_e$$

$$dH = TdS + Vdp + dw_e$$

$$dA = -SdT - pdV + dw_e$$

$$dG = Vdp - SdT + dw_e$$

$$dS_{sys} + dS_{surr} \geq 0$$

$$d(G)_{P,T} \leq 0$$

$$(dG)_{p,T} = dw_e$$

E.M.F. work

$$dG = -SdT + pdV + dw_e$$

$$\Delta G_{P,T} \leq 0$$

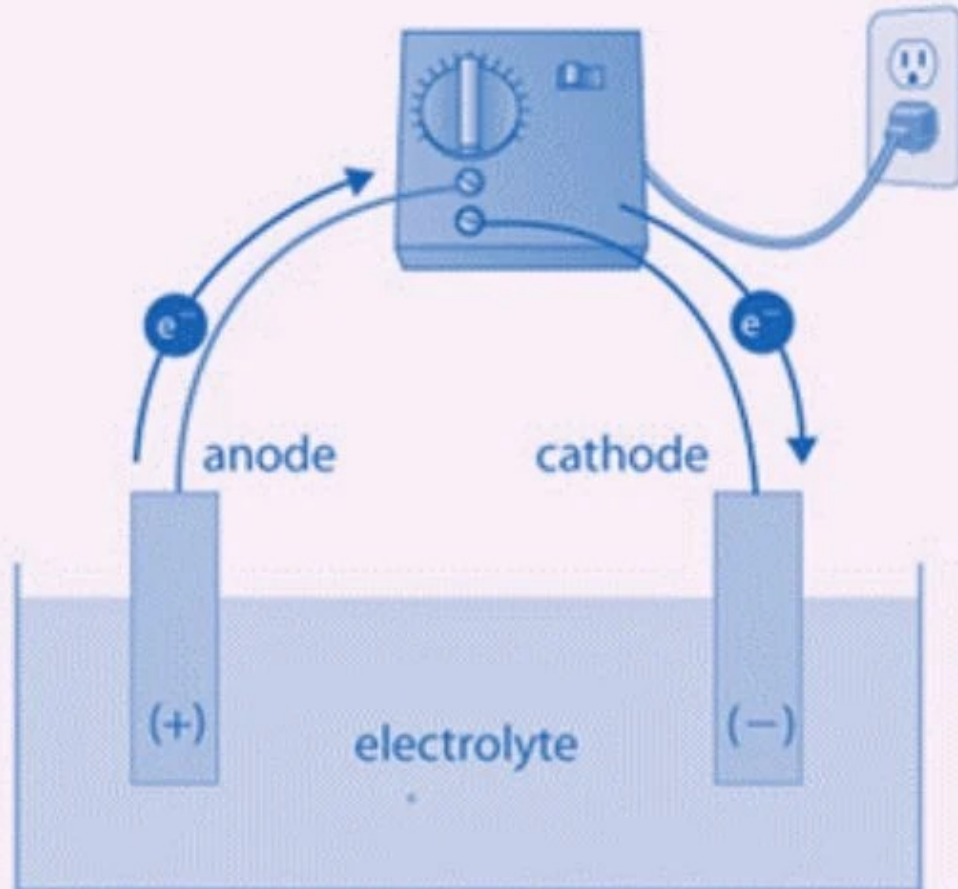
$$dG_{P,T} = dw_e = -nFdE$$

$$-nF\Delta E_{P,T} \leq 0$$

$$\Delta G_{P,T} = -nF\Delta E$$

$$\Delta E_{P,T} \geq 0$$

Electrolytic Cell



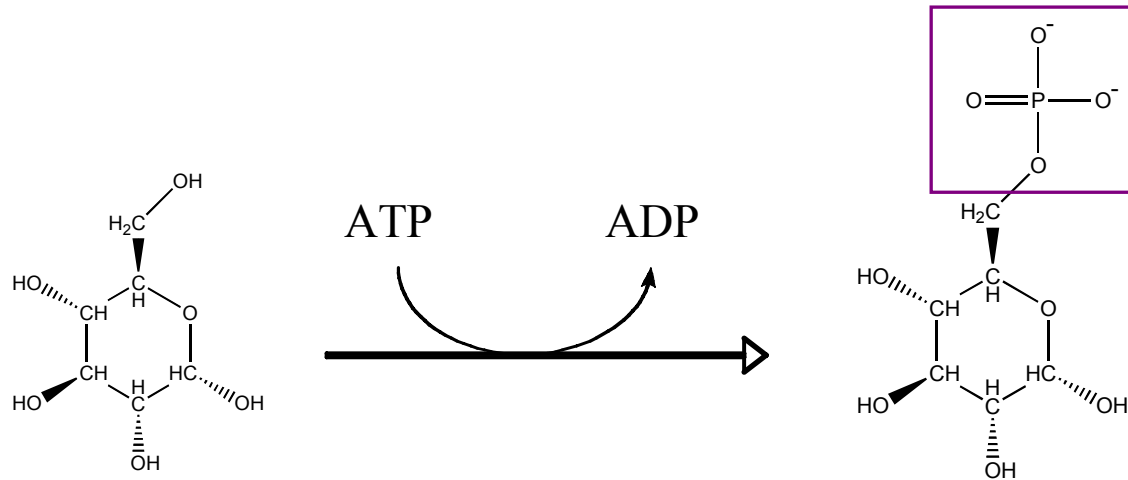
An electrolytic device that uses electrical energy to facilitate a non-spontaneous redox reaction is known as an electrolytic cell.

Coupling with a
spontaneous reaction



ATP hydrolysis as source for Driving Reaction

Reactants and products	ΔG° hydrolysis (kJ mol ⁻¹)
ATP + H ₂ O → ADP + P _i + H [⊕]	-32
ATP + H ₂ O → AMP + PP _i + H [⊕]	-45
AMP + H ₂ O → Adenosine + P _i + H [⊕]	-13
PP _i + H ₂ O → 2P _i	-29



$\Delta G^\circ = +17 \text{ kJ/mol}$ for glucose to glucose-6-P

$\Delta G^\circ = -31 \text{ kJ/mol}$ for ATP hydrolysis

An enzyme **couples** the glucose to glucose-6-P reaction to ATP hydrolysis.
 The net change in free energy = $17 \text{ kJ/mol} + (-31 \text{ kJ/mol}) = -14 \text{ kJ/mol}$

Formation of Ammonia

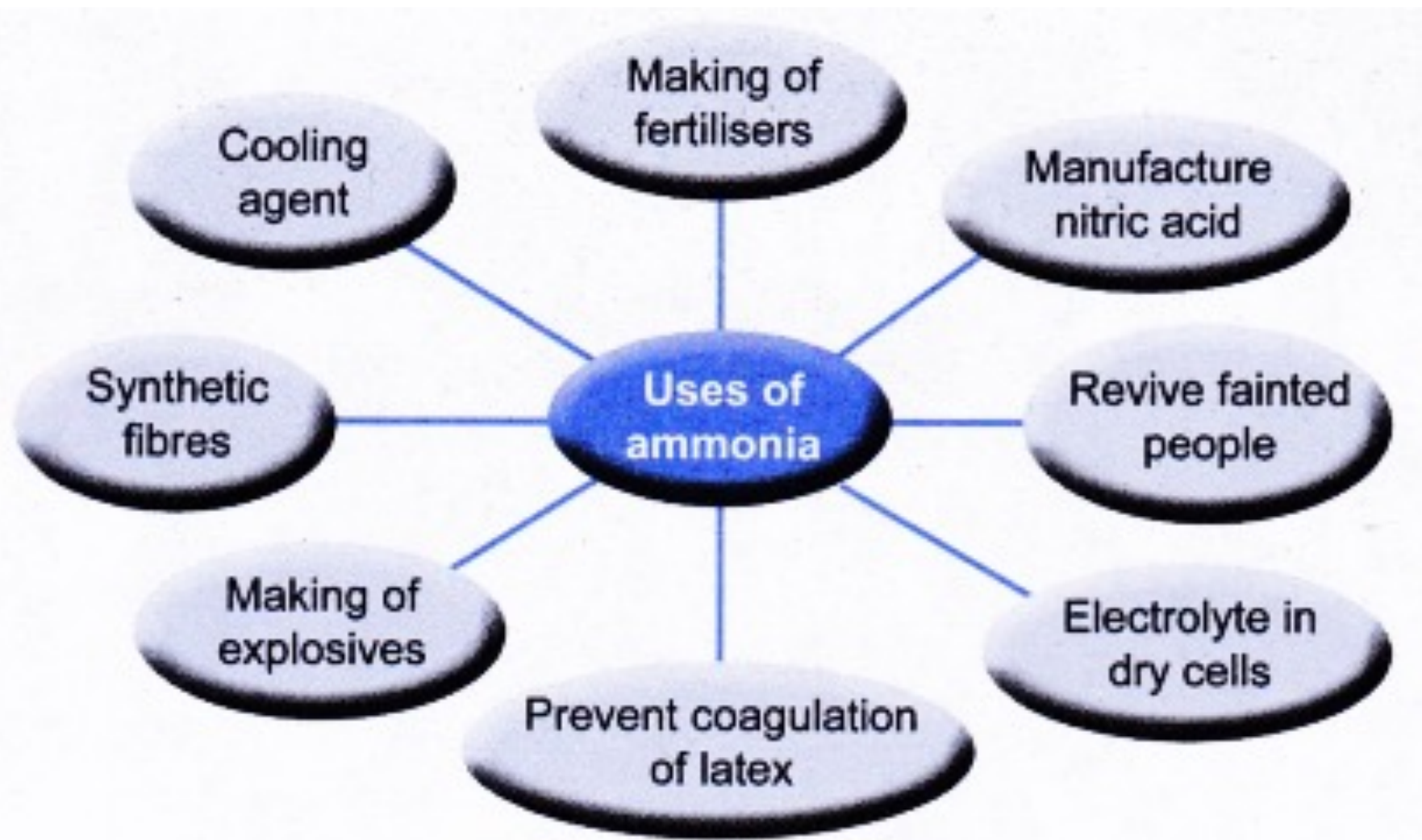
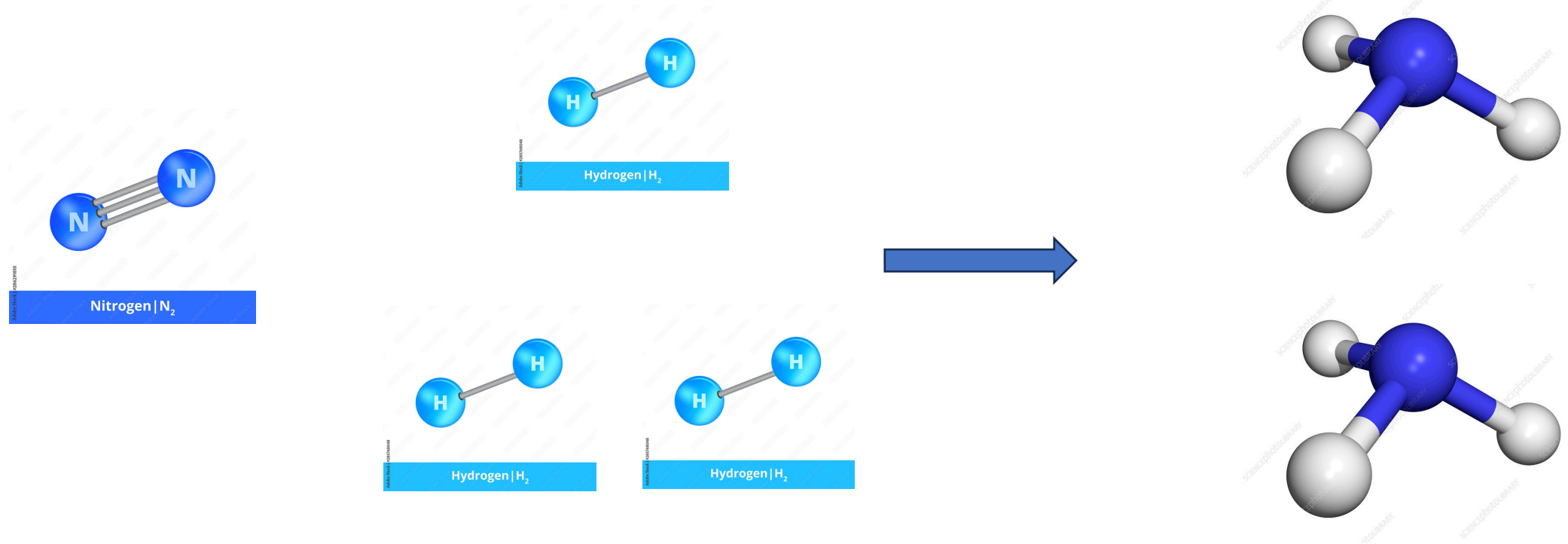


Figure *Uses of ammonia*

Balanced Reaction



1 mol of nitrogen gas will react with 3 mol of hydrogen gas to produce 2 mol of ammonia

28 g of nitrogen gas will react with 6 g of hydrogen gas to produce 34 g of ammonia

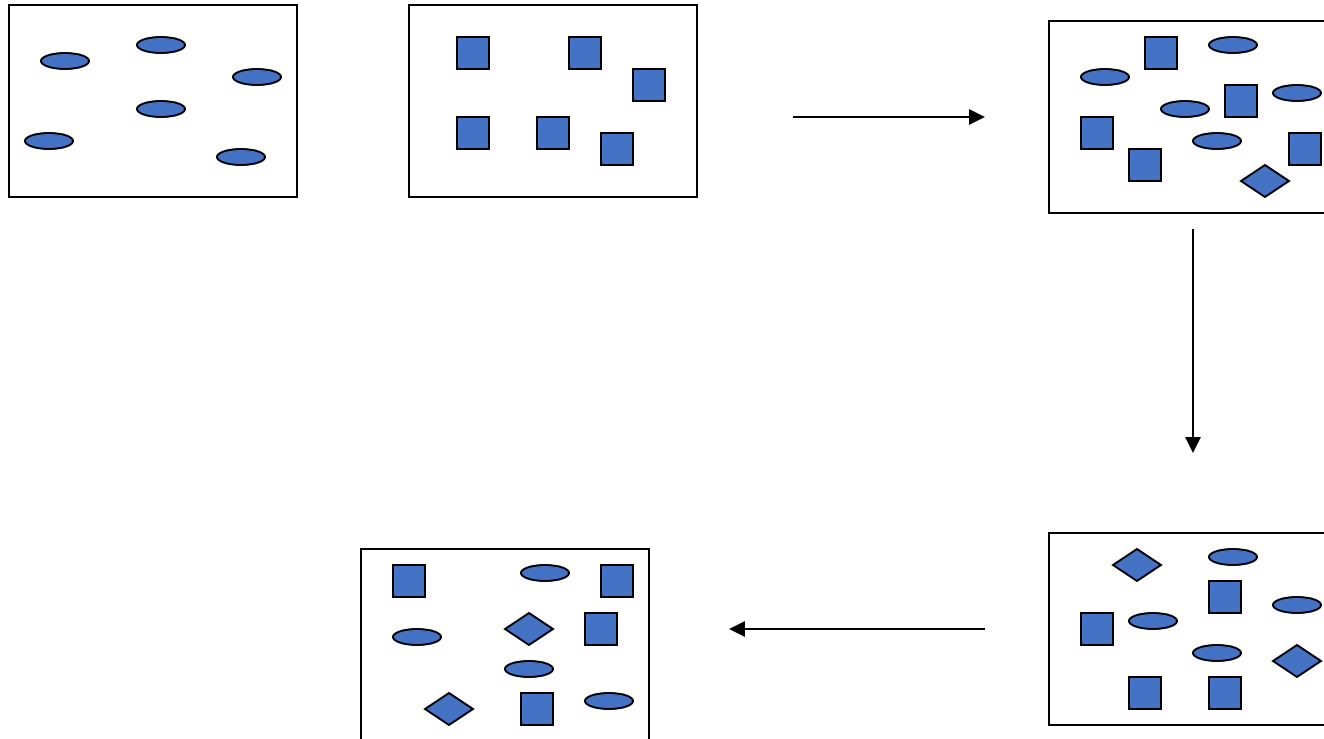
Nitrogen Fixation

- Microorganisms that fix nitrogen require 16 moles of adenosine triphosphate (ATP) to reduce each mole of nitrogen (Hubbell & Kidder, 2009).

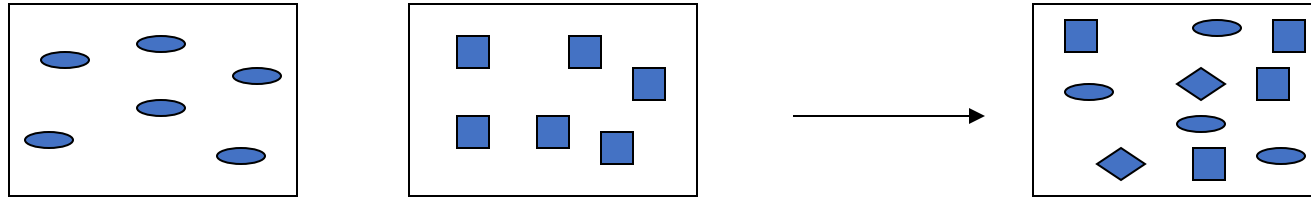


Chemical equilibrium

Equilibrium



Equilibrium



- All measurable properties of system remain constant over time.
- The rate of forward process at equilibrium is equal to rate of backward process.

Equilibrium Constant



**Rate of Forward Reaction=Rate of backward
Reaction**

$$k_f[\text{N}_2][\text{H}_2]^3=k_b[\text{NH}_3]^2$$

$$K = \frac{k_f}{k_b} = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$$

Equilibrium Constant

Types of Equilibrium Constant for a Chemical Process

There are three types of equilibrium constants, in which activity is expressed in terms of concentration (c), partial pressure (p), and mole fraction (x) of reactants and products.

For a chemical reaction, $aA + bB \rightleftharpoons cC + dD$,

$$K_c = \frac{[C]_{\text{eq}}^c [D]_{\text{eq}}^d}{[A]_{\text{eq}}^a [B]_{\text{eq}}^b}$$

$$K_p = \frac{[p_C]_{\text{eq}}^c [p_D]_{\text{eq}}^d}{[p_A]_{\text{eq}}^a [p_B]_{\text{eq}}^b}$$

$$K_x = \frac{[x_C]_{\text{eq}}^c [x_D]_{\text{eq}}^d}{[x_A]_{\text{eq}}^a [x_B]_{\text{eq}}^b}$$



$$\ln \frac{K_{p,2}}{K_{p,1}} = \frac{\Delta H^0}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

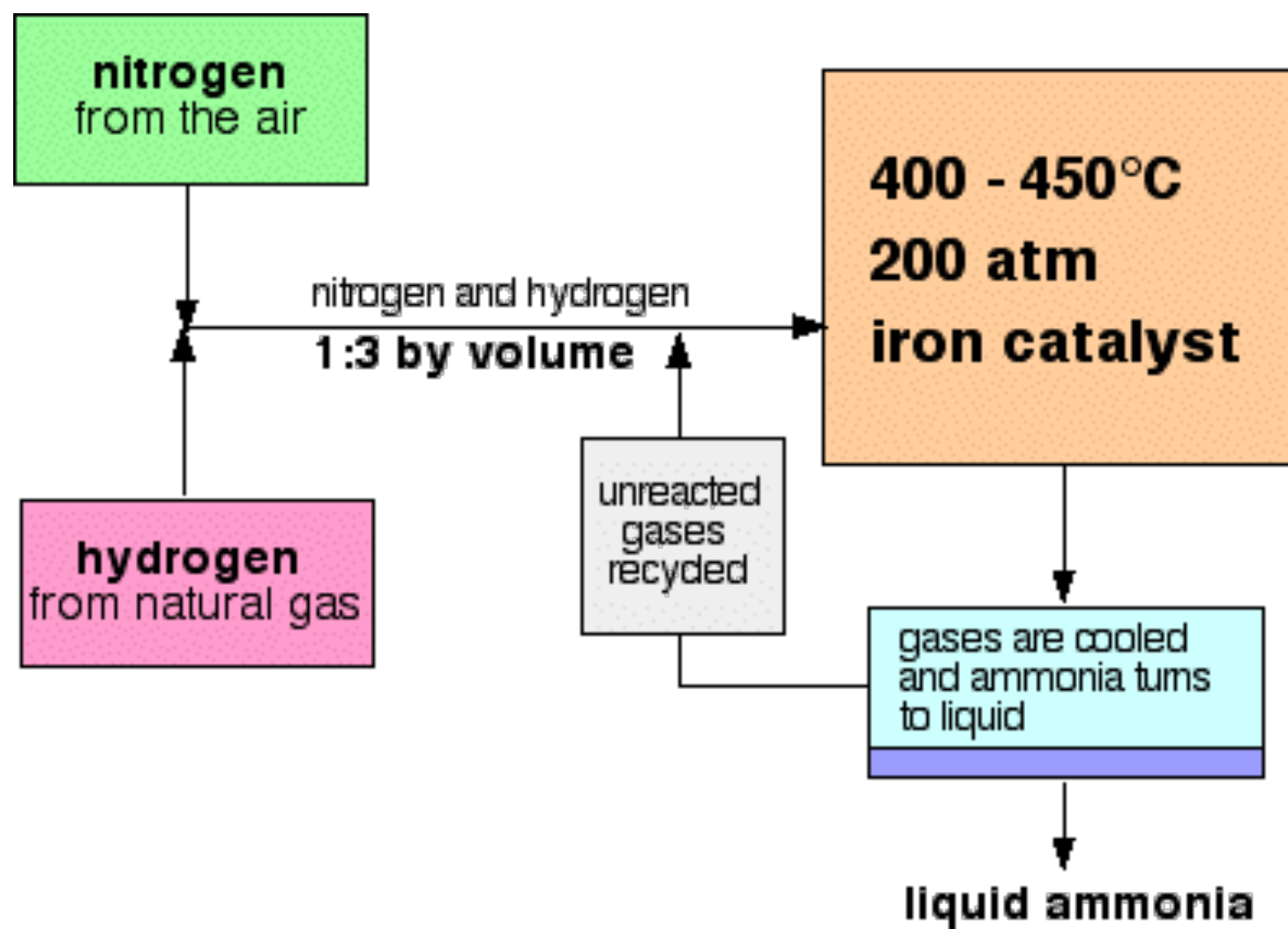


Temp	Enthalpy	dH/R	(T2-T1/T1*T2)	C*D	expE	K
300	-92000	-11065.672	2.23714E-05	-0.2475542	0.78070791	530881.38
350	-92000	-11065.672	0.000498562	-5.516922	0.0040182	2732.37397
400	-92000	-11065.672	0.000855705	-9.4689478	7.7213E-05	52.504572
450	-92000	-11065.672	0.001133482	-12.542746	3.5707E-06	2.42808377
500	-92000	-11065.672	0.001355705	-15.001784	3.0536E-07	0.20764281
550	-92000	-11065.672	0.001537523	-17.013724	4.0835E-08	0.02776785
600	-92000	-11065.672	0.001689038	-18.690341	7.6364E-09	0.00519274



$$K_p = \frac{p_{NH_3}^2}{p_{N_2} p_{H_2}^3} = \frac{x_{NH_3}^2}{x_{N_2} x_{H_2}^3} \times \frac{1}{p^2} = \frac{n_{NH_3}^2}{n_{N_2} n_{H_2}^3} \times \left(\frac{n_T}{p}\right)^2$$

NH3	N2 (1 mol)	H2 (3 mol)	Total mole	Kn	Kx	Kp at 1 atm	Kp at 10 atm
0.01	0.995	2.985	3.99	3.77871E-06	6.01575E-05	6.01575E-05	6.01575E-07
0.02	0.99	2.97	3.98	1.54225E-05	0.000244299	0.000244299	2.44299E-06
0.04	0.98	2.94	3.96	6.42469E-05	0.001007494	0.001007494	1.00749E-05
0.06	0.97	2.91	3.94	0.000150609	0.002337999	0.002337999	2.338E-05
0.08	0.96	2.88	3.92	0.000279082	0.00428848	0.00428848	4.28848E-05
0.1	0.95	2.85	3.9	0.000454718	0.006916256	0.006916256	6.91626E-05
0.2	0.9	2.7	3.8	0.002258012	0.032605689	0.032605689	0.000326057
0.3	0.85	2.55	3.7	0.00638562	0.087419132	0.087419132	0.000874191
0.4	0.8	2.4	3.6	0.014467593	0.1875	0.1875	0.001875
0.5	0.75	2.25	3.5	0.029263832	0.358481939	0.358481939	0.003584819
0.6	0.7	2.1	3.4	0.055532417	0.641954741	0.641954741	0.006419547
0.7	0.65	1.95	3.3	0.101666738	1.107150777	1.107150777	0.011071508
0.8	0.6	1.8	3.2	0.182898948	1.872885231	1.872885231	0.018728852
0.9	0.55	1.65	3.1	0.327846459	3.150604467	3.150604467	0.031506045

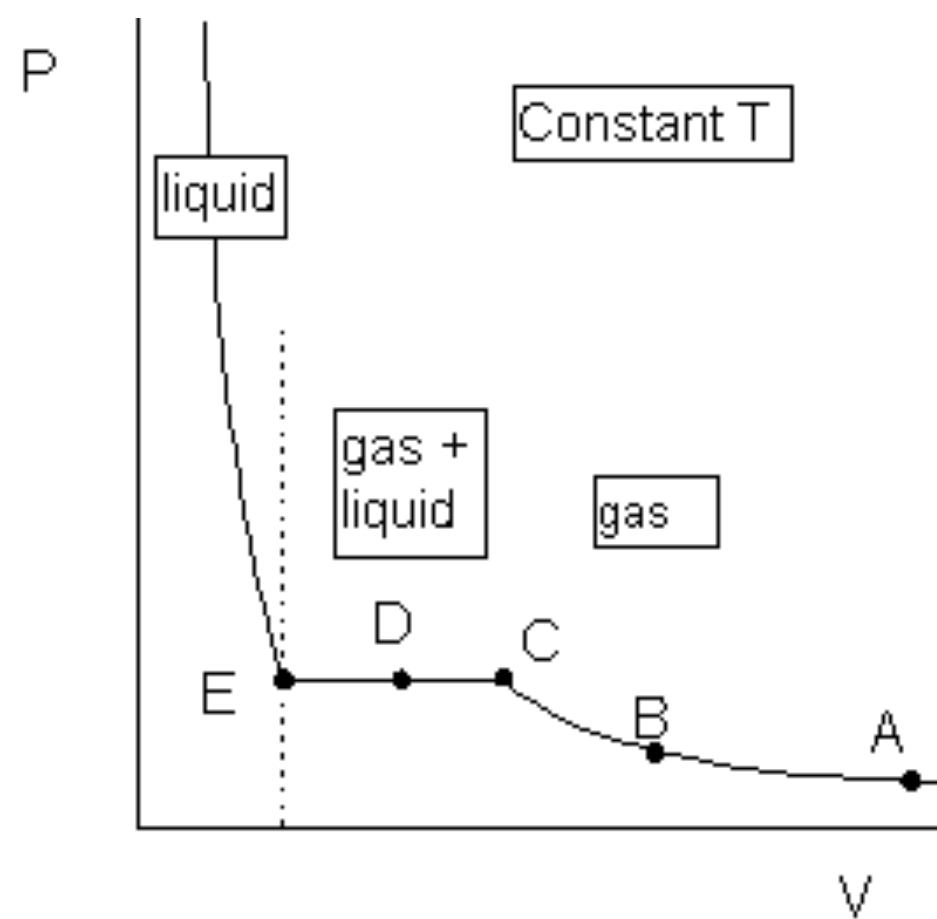


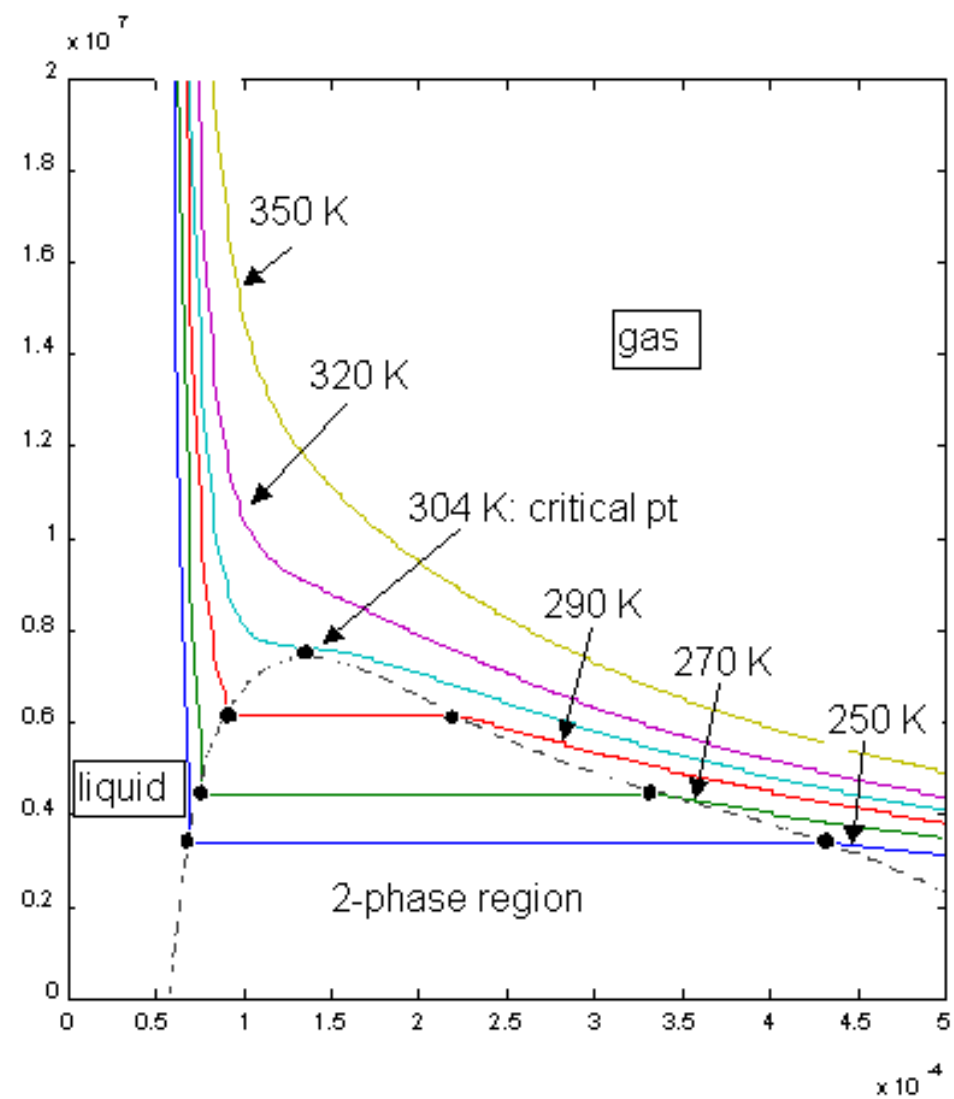


Fritz Haber
Nobel Prize, 1918



Carl Bosch
Nobel Prize, 1931





Critical constants of some gases

- The very low critical pressure and temperature of **helium**, reflecting the very small intermolecular attractions of this atom. • T_c of the **noble gas elements** increases with atomic number.
- **Hydrogen** gas cannot be liquified above 33 K; this poses a major difficulty in the use of hydrogen as an automotive fuel; storage as a high-pressure gas requires heavy steel containers which add greatly to its effective weight-per-joule of energy storage.
- The properties of **carbon dioxide** (particularly its use as a supercritical fluid).
- The high T_c of **H₂O** is another manifestation of its "anomalous" properties relating to hydrogen-bonding.

Chemical kinetics

Importance of kinetics

- Amount of product or intermediate or reactant at a given time.
- Stability and rate of drug metabolism: Dose and expiry date
- Degradation/decomposition rate.
- Slow or fast reaction.
- Radiometric dating.

Slow and Fast reactions

- This can be used to our advantage:
- The degradation of paint should be slow.
- Rusting of Iron need to be slowed down.
- Burning of cooking gas should be fast
- The rate of oxidation of a fuel should be negligible at room temperature, but it should be fast at the high temperature at which engine operates.

Importance of Kinetics

- Mechanism of reaction
- Effect of solvent on the reaction.
- Effect of substrate on the reaction.
- Effect of base/nucleophile/ electrophile on the reaction.

Importance of Catalysis

- Homogeneous catalysis
 - Acid-Base Catalysis
 - Enzyme Catalysis
- Heterogeneous catalysis